The Direct and Indirect Effects of Genetics and Education

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Work in Progress

Abstract

Recent advances in genetic technology measure genetic locations in the human genome associated with social outcomes, such as years of education, generating genetic measure called polygenic indices. I show how that the Education PolyGenic Index (Ed PGI, Lee et al. 2018) is related to later-life earnings, using random deviation away from parents' genes to infer causal effects. A nascent social science literature has speculated that genes associated with education relate to intelligence, and so would have independent, direct genetic effects on labour market outcomes. I investigate this systematically by using mediation methods to decompose the causal effects into a direct genetic effect, and an indirect effect from returns to education, using genetic and labour market data for thousands of members of the British public in the UK Biobank. I derive selection bias terms for conventional mediation methods when education is not randomly assigned — as is likely for any mediator in any natural experiment setting. To avoid this, I use a legal change in school-leaving age to instrument for random variation in education, estimating direct and indirect effects local to these compliers. Correlational evidence with public data shows that direct genetic associations are indistinguishable from zero — all causal associations of the Ed PGI are explained by the indirect education channel, and not by a direct genetic channel. Causal estimates using restricted UK Biobank data are to follow.

Keywords: Education, Mediation, Polygenic indicies, Nature, Nurture.

JEL Codes: D31, D91, I24, J24, Z00.

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> The difference of natural talents in different men, is, in reality, much less than we are aware of [...]. The difference between the most dissimilar characters, between a philosopher and a common street porter, seems to arise not so much from nature, as from habit, custom, and education.
>
> — *Smith (1776), The Wealth of Nations, Volume 1 p. 16.*

0.1 Plan

1. Refine the randomisation argument for Ed PGI. Using only the one copy gene variants from each parent, and comparing the results to previous research (i.e., Carvhalo 2024).
2. Structural model for the mediation analysis of Ed PGI, using the Roy model and control function identification to attain identification.

Section map for this paper:

1. Intro (done last).
2. Data + biological context.
3. Section on the adjusted PGI randomisation design, clearly stating estimands
4. New results of genetic effects.
5. Conceptual framework for mediation effects, clearly stating estimands
6. Results of the mediation analysis.
7. Conclusion.

Note: a major concern for the Ed PGI previous papers is that they have little economic usage. This paper can be framed as changing that, investigating the economic system underlying the Ed PGI causal effects, giving economic credence where before there was merely concern.

This paper proceeds as follows. Section 2 describes the UKB, including labour market and genetic data, and the genetic context for Mendelian inheritance. **??** shows results of the

first-stage Mendelian inheritance approach, and causal effects of the PGI on labour market outcomes. **??** gives a conceptual framework for estimating the direct and indirect effects of genetics and education, in this observational setting; Section 5 gives the empirical results. **??** concludes.

1 Introduction

Re-write with this, https://blogs.ubc.ca/khead/research/research-advice/formula.

Recent advances in genetic technology have allowed researchers to identify and measure genetic locations in the human genome associated with health and social outcomes, such as education attainment. These measures are genetic scores, and a nascent economics literature documents how the education attainment genetic score (Ed PGI, Lee et al. 2018) is related to many outcomes, including retirement wealth, life-cycle income, elementary school performance, and longevity (Barth et al. 2020, Marioni et al. 2016, Papageorge & Thom 2020, Houmark et al. 2022). This project reanalyses claims that the Ed PGI is associated with labour market outcomes, focusing on later-life labour earnings, through a direct genetic channel and indirect channel through take-up and returns to education.

I use data from the Health and Retirement Study (HRS) to estimate the correlation between Ed PGI, education, and later life earnings among $5,957$ retirement-age Americans. A one standard deviation increase in Ed PGI is associated with $+0.67$ years of education, $+7.2\%$ likelihood attending university, and $+5.2\%$ annual earnings. A higher Ed PGI could lead to better labour market outcomes by inducing people to attend more years of education (and education returns via this channel), but it could be associated thanks to a direct genetic channel (independent of education). I employ Causal Mediation (CM) methods to decompose the association of Ed PGI $\rightarrow$ earnings into an indirect (education) and direct path (absent education). CM estimates in HRS data show that around 40–60% of the association of Ed

PGI to earnings operates via increased education attainment — i.e., around half of the effect is because they attended more years of education.

Conventional CM methods rely on a selection-on-observables assumption, which may not hold true in observational work. I explicitly connect the assumptions behind CM methods to those of selection into treatment in classic labour and observation economic research (Heckman & Vytlacil 2005). When a mediator, here education, is not randomly assigned then conventional CM methods for estimating direct and indirect effects are contaminated by selection bias. I write this as both a non-parametric non-identification result, and with a model-based regression framework with a correlated error term(e.g., as in the Imai et al. 2010 linear model approach). Structural assumptions could solve the identification problem, for example if selection into education follows a Roy model or errors terms have a known distribution (Heckman 1979). Adjusting indirect and direct estimates with sample selection models gives estimates for the direct genetic channel indistinguishable from zero, assigning roughly all the association for Ed PGI to earnings by the education channel.

This work adds to a growing literature of genetics in economics (Barth et al. 2020), and expands on mediation methods (Imai et al. 2010) which are rarely used in empirical economic research (Huber 2020). The most similar papers have studied the association between Ed PGI and earnings (Papageorge & Thom 2020), and socioeconomic status (Carvalho 2024). Another has considered a similar topic from the view of genes as instruments, when the exclusion restriction is violated (Spiller et al. 2019, Van Kippersluis & Rietveld 2018). To the best of my knowledge, this is the first work to explicitly decompose direct and indirect genetic associations, and the first to connect mediation methods (and its selection-on-observables assumptions) to the labour economics literature for selection into treatment.

This paper proceeds as follows. Section 2 describes the HRS, including labour market and genetic data, and the biological context behind Mendelian randomisation. **??** presents the conceptual framework for estimating treatment effects when the genetic scores may affect

outcomes independently (without the exclusion restriction). Section 5 presents empirical results for direct and indirect effects. **??** concludes, with a discussion of the future steps and work in progress.

2 Data and Genetic Context

2.1 Health and Retirement Study (HRS)

The Health and Retirement Study (HRS) is a longitudinal survey that studies the households of American adults over 50 years old, measuring health and economic outcomes for individuals and their family every second year from 1992, with most recent data available in 2020.[1] I define the main analysis sample as household heads who provided genetic information, and who have non-missing measures on education attainment, and self-reported measures of annual income. These data form a panel for individuals, where each row is a household-head-year; from this panel, I take the maximum self-reported annual earnings at ages 50–65 (among those working), record the year this is reported in, the maximum years of education reported, and use this as the main analysis sample. I then restrict to individuals with race defined as "White," as the genetic measures are calculated among populations of European ancestry (see *Health and Retirement Study Polygenic Scores – Release 4.3* (2023) for further details).

There are $5,957$ individuals in this sample, a representative sample of retirement age, American household heads. Additionally, the HRS data include additional characteristics, including count of children, education years of their parents,[2] and the US region they reside in. I include these values in the analysis sample, taking values reported in the same year of the maximum annual earnings. Table 1 reports summary statistics for this sample, and

[1] The *Health and Retirement Study Core Public Use Dataset* (2020) is sponsored by the National Institute on Ageing (grant number NIA U01AG009740) and is conducted by the University of Michigan. I use the RAND corporation's compilation of HRS data.

[2] I take the maximum between mother and father's years of education, and report this value thanks to missing values.

Table 1: HRS Summary Statistics.

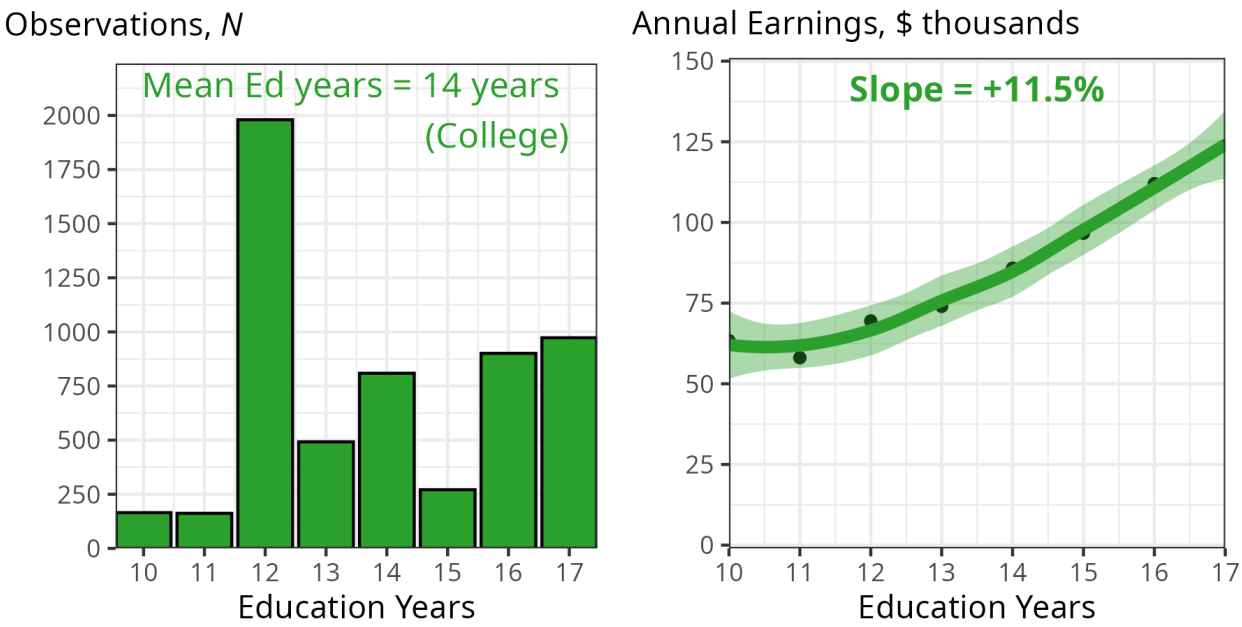
Statistic	Mean	St. Dev.	N
Ed PGI, continuous	0.01	1.00	5,957
Ed PGI, binary ¿ 0	0.51	0.50	5,957
Education Years	13.69	2.35	5,957
Higher Education Degree?	0.31	0.46	5,957
Annual Labour Earnings, $ thousands	87.69	56.04	5,957
Annual Total Income, $ thousands	149.45	72.54	5,957
Gender – Female	0.50	0.50	5,957
Household size	2.86	1.32	5,957
Count of Children	2.94	1.88	5,957
Parents' Education Years	11.82	3.00	5,957
Age During Survey	54.25	3.42	5,957
Survey Year	1,999.93	7.70	5,957
Survey Year, in 1990s?	0.55	0.50	5,957
Survey Year, in 2000s?	0.24	0.43	5,957
Survey Year, in 2010s?	0.22	0.41	5,957
USA Region – New England	0.05	0.21	5,957
USA Region – Atlantic	0.33	0.47	5,957
USA Region – Central	0.44	0.50	5,957
USA Region – Mountain	0.06	0.24	5,957
USA Region – Pacific	0.12	0.33	5,957
Years in Survey	3.62	1.94	5,957
Childhood SES: Family poor	0.23	0.42	5,957
Childhood SES: Family moved	0.15	0.35	5,957
Childhood SES: Family financial help	0.14	0.35	5,957
Childhood SES: Father missing	0.04	0.21	5,957
Childhood SES: Father unemployed	0.17	0.38	5,957
Childhood SES: Father manual labourer	0.53	0.50	5,957
Childhood SES: Bad childhood health	0.01	0.09	5,957
Childhood SES: Parents smoked	0.73	0.45	5,957
Childhood SES: Childhood head injury	0.12	0.32	5,957

Note: This table shows the summary statistics for the HRS analysis sample, household heads with sample restrictions described above.

Figure 1 summarises the correlation between education years and my measure of later-life labour earnings.

Figure 1: Distribution of HRS Education Measures.

(a) Education Years. **(b)** Education Years + Earnings.



Note: This figure shows the marginal distribution of education years in the HRS analysis sample, and the correlation between years of education and resulting earnings. Earnings represent the annual income of household heads, adjusted to 2022 by the CPI-U.

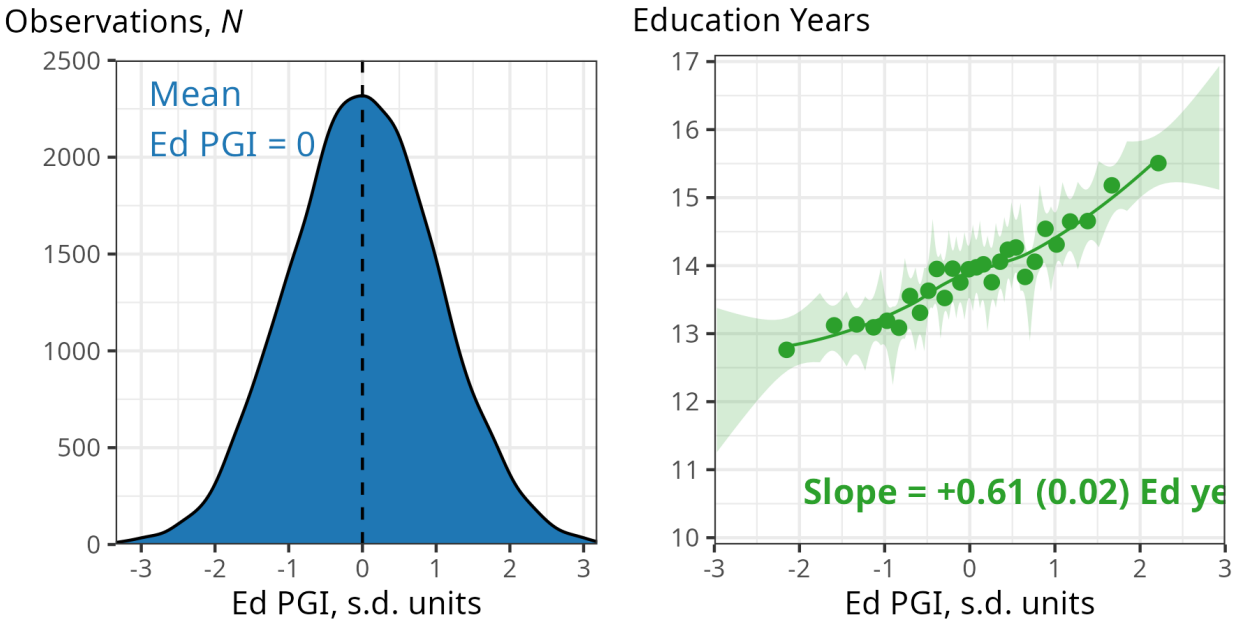
2.2 Education Polygenic Score (Ed PGI)

The HRS collected genetic information from survey respondents in the survey years 2006, 2008, 2010, 2012. This involves collecting respondents' saliva, and then measuring the respondent's genotype. From these genetic data, HRS then calculates genetic score for a number of individual outcomes (*Health and Retirement Study Polygenic Scores – Release 4.3* 2023). A genetic score is a measure of genetic predisposition for a certain trait (e.g., years of education, height, bipolar disorder, etc.), calculated from a complicated process. To summarise this process, biostatisticians measure outcomes and the full genome of thousands of people (mostly from biobank data), and identify the locations of single-letter changes on the human genome that predict (out-of-sample) the outcome of interest, calculating a set of weights for the gene locations. The gene score is then calculated (out-of-sample) for the

HRS participants, as the weighted sum of gene mutations that each HRS participant exhibits. Notably, most gene scores (and all included in this analysis) are calculated from populations of European ancestry, and are only applicable this population. See Uffelmann et al. (2021) for a full explanation of the approach to calculating polygenic scores.

Figure 2: Distribution of HRS Ed PGI.

(a) Ed PGI. **(b)** Ed PGI + Education Years.



Note: This figure shows the marginal distribution of Ed PGI and education years among HRS household heads, in the analysis sample defined above. Ed PGI is normalised within-in sample by the HRS to have mean 0, and standard deviation 1; the distribution shape is unaffected by the normalisation — i.e., it is roughly normally distributed regardless. 3.4% of the analysis sample reports education years less than 10 years (not included in panel b).

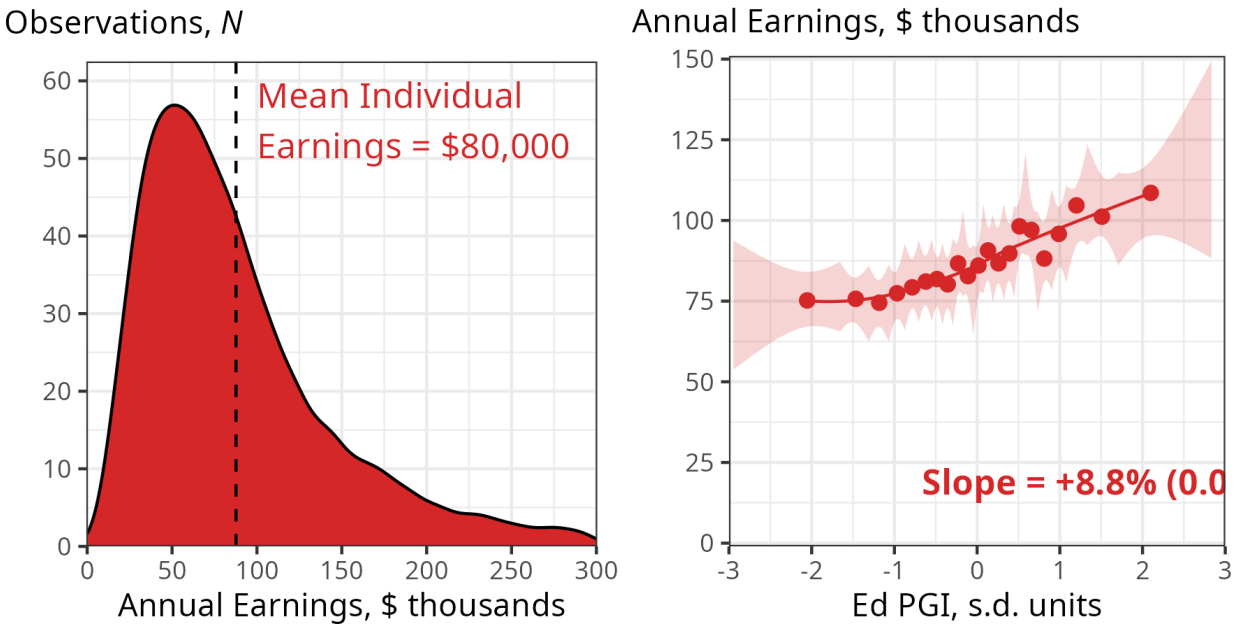
The HRS includes a gene score for education attainment (Ed PGI), a measure of genetic predisposition to years of education. The Ed PGI is calculated from weights developed by Lee et al. (2018) from genetic and education data from over a million individuals' genetic data, from the UK Biobank and other biobanks. Figure 2 shows the marginal distributions of the Ed PGI and education years among the sample of $5,957$ HRS household heads. Ed PGI is normalised with mean 0 and standard deviation 1, and is roughly normally distributed among the HRS sample; years of education is bunched, with around 40% of the sample having

achieved at most high school (12 years), while 32% achieve a college degree.

Epidemiology has a recent history of using these gene scores as instruments for health conditions, an approach known as Mendelian Randomisation (MR, Lawlor et al. 2008). The key reasoning is that at meiosis there is a random assortment of parents genes mixing (Mendelian independent assortment), so that a child inherits a random mix of genes from their parents and not a pre-determined 50-50 mix of genes from each parent. The HRS does not have measures of parents' genotype, or many siblings, so following Papageorge & Thom (2020) I condition on the measure of parents' education through-out. This means that the approach in this version of the paper is decomposing genetic *associations*, and future work will exploit the causal framework to decompose *causal effects*.

Figure 3: Distribution of HRS Earnings.

(a) Earnings. **(b)** Ed PGI + Earnings.



Note: This figure shows the marginal distribution of Ed PGI and education years among HRS household heads, in the analysis sample defined above. Ed PGI is normalised within-in sample by the HRS to have mean 0, and standard deviation 1; the distribution shape is unaffected by the normalisation — i.e., it is roughly normally distributed regardless. 3.4% of the analysis sample reports education years less than 10 years (not included in panel b).

The key Lee et al. (2018) finding is replicated in Figure 3: Ed PGI predicts 10–12% of

the variation in education years. This graphically represents the first-stage estimates crucial in MR studies, where the first-stage is very strong.[3]

The Ed PGI is calculated from 1,271 different genes (Lee et al. 2018), each of which are found to significantly predict year of education. Papageorge & Thom (2020) document how the Ed PGI predicts earnings, even when controlling for education, and posit that the score contains genes that affect skills related to financial literacy.

3 Mendelian Independent Inheritance

Genes are not randomly assigned to children. Genes are inherited from parents, where children receive 13 chromosomes from each parent at conception — half the genes from their mother, half from father.

Parents' traits, such their socioeconomic status, may differ for parents with higher or lower PGI values, confounding association-based estimates of PGI genetic effects among their children.

Causal mediation decomposes causal effects into two channels, through a mediator (indirect effect) and through all other paths (direct effect).

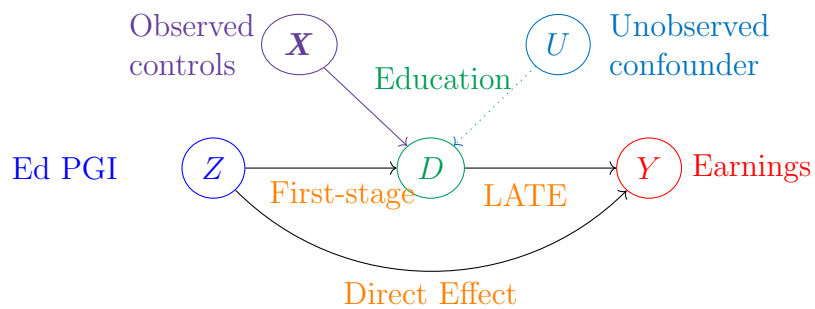
To develop notation for direct and indirect effects, write Z_i for an exogenous binary variable, D_i an intermediary outcome, and Y_i an outcome for individuals $i = 1, \ldots, n$. The outcomes are a sum of their potential outcomes:

$$D_i = Z_i D_i(1) + (1 - Z_i) D_i(0),$$
$$Y_i = Z_i Y_i(1, D_i(1)) + (1 - Z_i) Y_i(0, D_i(0))$$

[3]Note that the Ed PGI was constructed from UK Biobank data, yet predicts 10–12% variation in years of education in the the HRS data, out-of-sample.

Z_i affects outcome Y_i directly, and indirectly via the $D_i(Z_i)$ channel, with no reverse causality. The framework is general to any (conditionally) randomly assigned Z_i, selected mediator D_i, and outcome Y_i. For the analysis in Section 2, Z_i is the Ed PGI, D_i a measure of education year, and Y_i (log) later-life earnings. Note that Z_i, D_i are continuous measures in HRS Data, but this section focuses on binary $Z_i, D_i = 0, 1$ to simplify the causal framework.[4] Figure 4 visualises the design, where the direction arrows denote the causal direction (and no reverse causality).

Figure 4: Structural Causal Model for Direct and Indirect Effects of Genetics and Education.



Note: This figures shows the structural causal model for decomposing the direct and indirect effects of genetics and education attainment.

Assuming that Ed PGI Z_i is randomly assigned (or conditionally so, see Subsection 2.2), then there are only two average effects which are identified. The first-stage effect refers to the effect of the Ed PGI on education, $Z \to D$.

$$\mathbb{E}\left[D_i \mid Z_i = 1\right] - \mathbb{E}\left[D_i \mid Z_i = 0\right] = \mathbb{E}\left[D_i(1) - D_i(0)\right]$$

It common in the economics literature to assume that Z influences D in at most one direction, $\Pr\left(D_i(1) \geq D_i(0)\right) = 1$ — monotonicity (Imbens & Angrist 1994). I assume monotonicity (and its conditional variant) holds through-out, as it brings the mediation notation closer to the IV literature from labour economics.[5]

[4]Continuous analogues of the following are extensions of the binary case, and will be included in work to follow.

[5]Monotonicity has other beneficial implications in this setting, as shown in Subsection 4.1.

The reduced-form effect refers to the effect of the Ed PGI on earnings, $Z \to Y$, and is also known as the intent-to-treat effect in experimental settings, or total effect in causal mediation literature.

$$\mathbb{E}\left[Y_i \mid Z_i = 1\right] - \mathbb{E}\left[Y_i \mid Z_i = 0\right] = \mathbb{E}\left[Y_i(1, D_i(1)) - Y_i(0, D_i(0))\right]$$

On the other hand, mediation aims to decompose the reduced form effect of $Z \to Y$ into two separate pathways: indirectly through D, and directly absent D.

$$\begin{aligned} \text{Indirect Effect, } D(Z) \to Y: &\quad \mathbb{E}\left[Y_i(Z_i, D_i(1)) - Y_i(Z_i, D_i(0))\right] \\ \text{Direct Effect, } Z \to Y: &\quad \mathbb{E}\left[Y_i(1, D_i(Z_i)) - Y_i(0, D_i(Z_i))\right] \end{aligned}$$

These effects are not separately identified without further assumptions.

3.1 Causal Mediation Estimates

The conventional approach to estimating direct and indirect effects assumes both Z_i and D_i are conditionally ignorable.

Definition 1. *Sequential Ignorability (Imai et al. 2010).*

$$Z_i \perp\!\!\!\perp D_i(z), Y_i(z', d) \mid \boldsymbol{X}_i, \qquad \text{for } z, z', d = 0, 1 \tag{1}$$

$$D_i \perp\!\!\!\perp Y_i(z', d) \mid \boldsymbol{X}_i, Z_i = z', \qquad \text{for } z', d = 0, 1 \tag{2}$$

If 1(1) and 1(2) hold, then the direct and indirect effects are identified by two-stage mean differences, after conditioning on $\boldsymbol{X}_i$:[6]

[6] Imai et al. (2010) show a general identification statement; I show identification in terms of two-stage regression, which is more familiar in economics. This reasoning is in line with G-computation reasoning (Robins 1986); Subsection A.1 states the Imai et al. (2010) identification result, and then develops the two-stage regression notation which holds as a consequence of sequential ignorability.

$$\mathbb{E}_{D_i,\boldsymbol{X}_i}\left[\underbrace{\mathbb{E}\left[Y_i \mid Z_i=1, D_i, \boldsymbol{X}_i\right]-\mathbb{E}\left[Y_i \mid Z_i=0, D_i, \boldsymbol{X}_i\right]}_{\text{Second-stage regression, } Y_i \text{ on } Z_i \text{ holding } D_i \text{ constant}}\right]=\underbrace{\mathbb{E}\left[Y_i(1, D_i(Z_i))-Y_i(0, D_i(Z_i))\right]}_{\text{Average Direct effect}}$$

$$\mathbb{E}_{Z_i,\boldsymbol{X}_i}\left[\underbrace{\left(\mathbb{E}\left[D_i \mid Z_i=1, \boldsymbol{X}_i\right]-\mathbb{E}\left[D_i \mid Z_i=0, \boldsymbol{X}_i\right]\right)}_{\text{First-stage regression, } D_i \text{ on } Z_i} \times \underbrace{\left(\mathbb{E}\left[Y_i \mid Z_i, D_i=1, \boldsymbol{X}_i\right]-\mathbb{E}\left[Y_i \mid Z_i, D_i=0, \boldsymbol{X}_i\right]\right)}_{\text{Second-stage regression, } Y_i \text{ on } D_i \text{ holding } Z_i \text{ constant}}\right]$$

$$=\underbrace{\mathbb{E}\left[Y_i(Z_i, D_i(1))-Y_i(Z_i, D_i(0))\right]}_{\text{Average Indirect effect}}$$

These estimands are typically estimated with linear models (Imai et al. 2010):

$$D_i=\phi+\pi Z_i+\boldsymbol{\psi}_1^{\prime} \boldsymbol{X}_i+\eta_i$$

$$Y_i=\alpha+\beta D_i+\gamma Z_i+\delta Z_i D_i+\boldsymbol{\psi}_2^{\prime} \boldsymbol{X}_i+\varepsilon_i$$

And so the direct and indirect effects are composed from OLS estimates, $\widehat{\gamma}+\widehat{\delta}\mathbb{E}\left[D_i\right]$ for the direct effect and $\widehat{\pi}\left(\widehat{\beta}+\mathbb{E}\left[Z_i\right]\widehat{\delta}\right)$ for the indirect effect. While this is the most common approach in the applied literature, I do not focus on the linear formulation of this problem as it assumes homogenous treatment effects and linear confounding. These assumptions are unnecessary to my analysis; it suffices to note that heterogeneous treatment effects and non-linear confounding would bias OLS estimates of direct and indirect effects in the same manner that is well documented elsewhere (see e.g., Angrist 1998, Słoczyński 2022). I focus on fundamental problems that plague causal mediation methods in practice, regardless of estimation method. As such, I focus my work on non-parametric identification, and employ semi- and non-parametric estimation methods in my empirical analysis whenever possible to avoid these problems.

3.2 Selection Bias in Causal Mediation Estimates

Mediation methods are the main method that researchers then answer the following question: how did Z lead to a causal effect on Y, and through which channels? In observational work this may include a natural experiment that quasi-randomly assigns Z_i to individuals, regardless of their preferences or selection patterns — i.e., justifying assumption 1(1). Rarely does observational research employ an additional, overlapping identification design for D_i as part of the analysis, and instead they employ mediation methods by assuming this D_i is ignorable given observed covariates $\boldsymbol{X}_i$.[7] This approach leads to biased estimates, and contaminates inference regarding direct and indirect effects (in the same manner as Heckman et al. 1998).

Theorem 1. *Absent an identification strategy for the mediator, causal mediation estimates are at risk of selection bias. Suppose 1(1) holds, but 1(2) does not. Then causal mediation estimates are contaminated by selection bias terms, and group differences terms.*

Proof. See Subsection A.4 for the extended proof. □

Below I present the relevant selection bias and group difference terms, omitting the conditional on $\boldsymbol{X}_i$ notation for brevity. These selection bias terms would be equal to zero if the mediator was conditional ignorable (2), but do not necessarily average to zero if not.

For the average direct effect: CM estimate = ADE + selection bias + group differences.

[7] Imai et al. (2013) call attention to the need for a separate research design to isolate causal effects of D_i in randomised controlled trials; Subsection A.3 overviews literature, finding many papers that employ mediation methods with a research design for Z_i, but not for D_i.

$$
\begin{aligned}
&\mathbb{E}_{D_i}\Big[\mathbb{E}\left[Y_i \mid Z_i = 1, D_i\right] - \mathbb{E}\left[Y_i \mid Z_i = 0, D_i\right]\Big] \\
&= \mathbb{E}\left[Y_i(1, D_i(Z_i)) - Y_i(0, D_i(Z_i))\right] \\
&\quad + \mathbb{E}_{D_i}\Big[\mathbb{E}\left[Y_i(0, D_i(Z_i)) \mid D_i(1) = d\right] - \mathbb{E}\left[Y_i(0, D_i(Z_i)) \mid D_i(0) = d\right]\Big] \\
&\quad + \mathbb{E}_{D_i}\left[\Big(1 - \Pr\left(D_i(1) = d\right)\Big)\begin{pmatrix} \mathbb{E}\left[Y_i(1, D_i(Z_i)) - Y_i(0, D_i(Z_i)) \mid D_i(1) = d\right] \\ - \mathbb{E}\left[Y_i(1, D_i(Z_i)) - Y_i(0, D_i(Z_i)) \mid D_i(0) = 1 - d\right] \end{pmatrix}\right]
\end{aligned}
$$

For the average indirect effect: CM estimate = AIE + selection bias + group differences.

$$
\begin{aligned}
&\mathbb{E}_{Z_i}\left[\Big(\mathbb{E}\left[D_i \mid Z_i = 1\right] - \mathbb{E}\left[D_i \mid Z_i = 0\right]\Big) \times \Big(\mathbb{E}\left[Y_i \mid Z_i, D_i = 1\right] - \mathbb{E}\left[Y_i \mid Z_i, D_i = 0\right]\Big)\right] \\
&= \mathbb{E}\left[Y_i(Z_i, D_i(1)) - Y_i(Z_i, D_i(0))\right] \\
&\quad + \Pr\left(D_i(1) = 1, D_i(0) = 0\right)\Big(\mathbb{E}\left[Y_i(Z_i, 0) \mid D_i = 1\right] - \mathbb{E}\left[Y_i(Z_i, 0) \mid D_i = 0\right]\Big) \\
&\quad + \Pr\left(D_i(1) = 1, D_i(0) = 0\right) \times \\
&\quad \left[\begin{array}{l} \Big(1 - \Pr\left(D_i = 1\right)\Big)\begin{pmatrix} \mathbb{E}\left[Y_i(Z_i, 1) - Y_i(Z_i, 0) \mid D_i = 1\right] \\ - \mathbb{E}\left[Y_i(Z_i, 1) - Y_i(Z_i, 0) \mid D_i = 0\right] \end{pmatrix} \\ + \left(\dfrac{1 - \Pr\left(D_i(1) = 1, D_i(0) = 0\right)}{\Pr\left(D_i(1) = 1, D_i(0) = 0\right)}\right)\begin{pmatrix} \mathbb{E}\left[Y_i(Z_i, 1) - Y_i(Z_i, 0) \mid D_i(1) = 0 \text{ or } D_i(0) = 1\right] \\ - \mathbb{E}\left[Y_i(Z_i, 1) - Y_i(Z_i, 0)\right] \end{pmatrix} \end{array}\right]
\end{aligned}
$$

The selection bias terms come from systematic differences between the treated and untreated groups, differences not fully unexplained by $\boldsymbol{X}_i$. The group differences represent the fact that a matching estimator gives an average effect on the treated group and, when selection-on-observables does not hold, this is systematically different from the average effect (Heckman et al. 1998).[8] The group differences term is longer for the average indirect effect

[8]The selection-on-observables approach could, instead, focus on the average effect on treated populations (as do Keele et al. 2015). This runs into a problem of comparisons: CM estimates would give average effects on different treated groups. The CM estimate for the ADE on treated gives the ADE local to the $Z_i = 1$ treated group, while the AIE estimate gives the AIE local to the $D_i = 1$ group. In this way, these ADE and AIE on treated terms are not comparable to each other, so I focus on the true average terms.

estimate, because the indirect effect is comprised from the effect of D_i local to Z_i compliers; a matching estimator gets the average effect on treated, and the longer term adjusts for differences with the complier average effect.

4 Direct and Indirect Effects Under Selection

This section connects causal mediation, without assuming the mediator is randomly assigned (i.e., under selection), with classic labour economics models for selection into treatment.

4.1 Selection Model Representation

The IV literature assumes a first-stage monotonicity condition, where randomised Z_i influences mediator D_i in at most one direction.

Definition 2. *First-stage Monotonicity (Imbens & Angrist 1994).*

$$\Pr\left(D_i(1) \geq D_i(0)\right) = 1 \tag{3}$$

Assuming 2(3) in a mediation setting opens mediation to the wide literature on IV and selection models for identification in the presence of selection.

Theorem 2. *Under monotonicity, mediator D_i can be represented by a selection model. Suppose 2(3) holds, then there is a function $\mu(.)$ and random variable U_i such that D_i takes the following form.*

$$D_i(z) = \mathbb{1}\left\{\mu(z) \geq U_i\right\}, \quad \forall z = 0, 1$$

Proof. Special case of the Vytlacil (2002) equivalence result; see Subsection A.5. □

Theorem 2 is a powerful result: it says that at the cost of assuming monotonicity (as is

done in the IV literature), then selection into D_i takes a latent index form, and opens up identification in a mediation context to the wide literature on identifying treatment effects in selection models.

4.2 A Regression Framework for Direct and Indirect Effects

Inference for direct and direct effects can be written in a regression framework, showing how correlation between the error term and the mediator persistently biases estimates. And thus, selection models can be used to adjust for unobserved confounding.

To motivate a regression framework with unobserved confounding, write $Y_i(Z, D)$ as a sum of observed factors $Z_i, \boldsymbol{X}_i$ and unobserved factors, (following the notation of Heckman & Vytlacil 2005). First, define the following unobserved error terms

$$U_{0,i} = Y_i(Z_i, 0) - \mathbb{E}\left[Y_i(Z_i, 0) \mid \boldsymbol{X}\right], \quad U_{1,i} = Y_i(Z_i, 1) - \mathbb{E}\left[Y_i(Z_i, 1) \mid \boldsymbol{X}\right]$$

Then observed data take the following representation, which characterises direct effects, indirect effects, and the selection problem (see Subsection A.6 for all definitions).

$$\begin{aligned} D_i &= \phi_i + \pi_i Z_i + \eta_i \\ Y_i &= \alpha_i + \beta_i D_i + \gamma_i Z_i + \delta_i Z_i D_i + \underbrace{U_{0,i} + D_i\left(U_{1,i} - U_{0,i}\right)}_{\text{Correlated error term.}} \end{aligned}$$

And the average direct and indirect effects are given by the following

$$\begin{aligned} \mathbb{E}\left[Y_i(Z_i, D_i(1)) - Y_i(Z_i, D_i(0))\right] &= \mathbb{E}\left[(\beta_i + Z_i\delta_i) \times \pi_i\right], \\ \mathbb{E}\left[Y_i(1, D_i(Z_i)) - Y_i(0, D_i(Z_i))\right] &= \mathbb{E}\left[\gamma_i + \delta_i D_i\right]. \end{aligned}$$

By assumption $Z_i \perp\!\!\!\perp Y_i(.,.), D_i(.)$, so that the regression only gives unbiased estimates if

D_i is also conditionally random: $D_i \perp\!\!\!\perp U_{0,i} - U_{1,i} \mid \boldsymbol{X}_i$. If not, then the regression estimates (without adjusting for the contaminated bias term) suffer from omitted variables bias.

4.3 Selection into Education

In the education context, point identifying direct and indirect effects requires the *researcher controls for all sources of selection-into-education.*

While this assumption may hold true in two-way randomised experiments (e.g., in a laboratory or two-way RCT), it is unlikely to hold in the case of quasi-experimental variation in Z, or when modelling education as a mediator — absent a separate identification strategy for education D. To expand this point in an econometric selection-into-treatment framework, suppose selection follows a Roy model, where individual i weighs the costs and benefits of completing education.

$$D_i(Z_i) = \mathbb{1}\left\{\underbrace{C_i(Z_i)}_{\text{Costs}} \leq \underbrace{Y_i(Z_i, 1) - Y_i(Z_i, 0)}_{\text{Gains}}\right\}$$

Education choice $D_i(z)$ is clearly related to $Y_i(z, d)$ in this model, so let's see what the equation looks like in terms of sequential ignorability. As above, decompose costs into observed and unobserved factors.

$$C_i(Z_i) = \mu_C(Z_i; \boldsymbol{X}_i) + U_{C,i}$$

And so we can write the first-stage selection equation in full.

$$D_i(z) = \mathbb{1}\left\{\underbrace{U_{C,i} + U_{0,i} - U_{1,i}}_{\text{Unobserved}} \leq \underbrace{\mu_1(z; \boldsymbol{X}_i) - \mu_0(z; \boldsymbol{X}_i) - \mu_C(z; \boldsymbol{X}_i)}_{\text{Observed}}\right\}$$

Sequential ignorability, where $Y_i(z, d) \perp\!\!\!\perp D_i(z') \mid \boldsymbol{X}_i$, would then require that $\mathbb{E}\left[U_{0,i} - U_{1,i} \mid D_i\right] =$

0. In the Roy model above, this would assume every single contribution for returns to education is contained in $\boldsymbol{X}_i$; if there are any unobserved sources by which people have systematically different returns to education, then they would select into education based on this, and bias naïve mediation estimates. This is unlikely to hold true, unless there is another identification strategy for D_i — in addition to the one used for Z_i.

4.4 Estimating Direct and Indirect Effects

Quasi-experimental work does not take the assumption of "selection-on-observables" at face value without an explicit research design (Angrist & Pischke 2009) or modelling approach to address this issue.

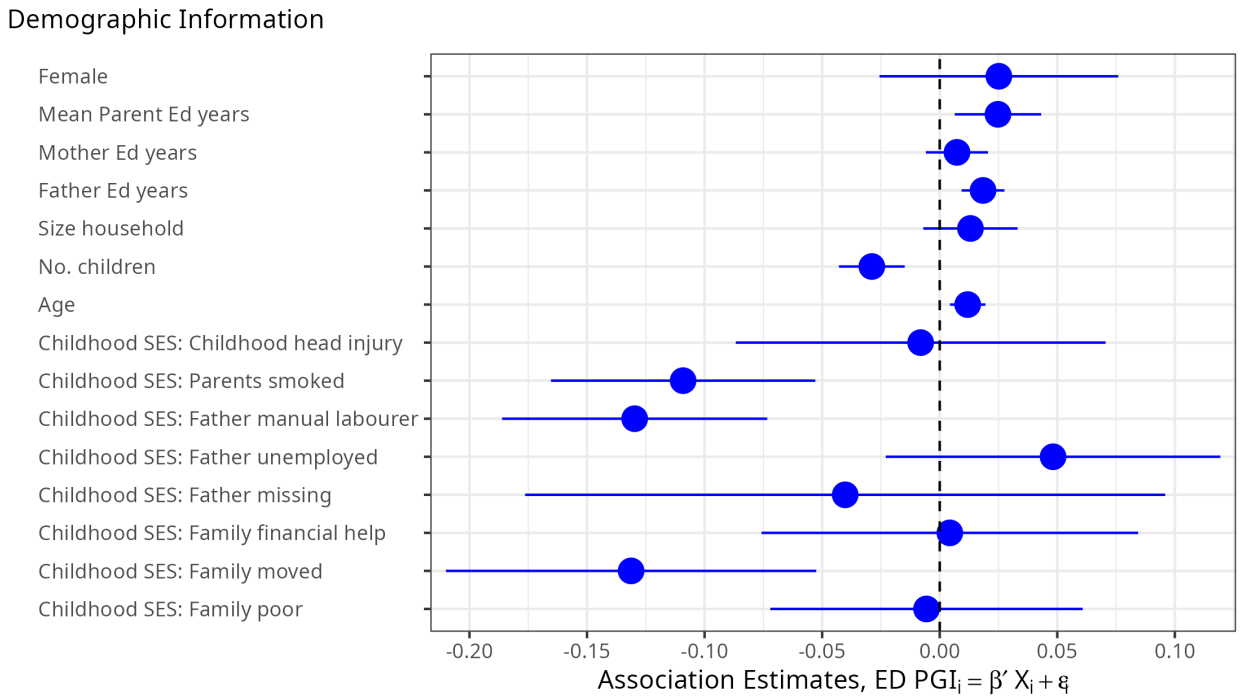
A classical approach to modelling this issue, is a selection model approach (Heckman 1974, 1979). The approach assume U_0, U_1 follow a known distribution (e.g, bivariate normal), and estimates the regression via maximum likelihood. Alternatively, a control function approach estimates the system in two stages, avoiding (some) distributional assumptions if an instrument is used, at the cost of efficiency. In the following, I estimate direct and indirect effects first by OLS (assuming sequential ignorability), and then via both variants of the sample selection models, to compare estimates. Future work will consider estimates by using an alternative instrument for education, in the framework of Frölich & Huber (2017) to avoid the modelling assumptions inherent to sample selection models.

5 Empirical Results

Before explicitly modelling direct and indirect effects, I first show estimates of a familiar and conventional regression estimates for between Ed PGI and demographics. Ed PGI is correlated with years of parents' education, where those born to more educated parents have a higher Ed PGI on average. This reflects heritability of Ed PGI, and exhibits why

it is important to include years of parents' education as an explicit control in this analysis. Similarly, for some measures of early childhood socioeconomic status, where those with lower Ed PGIs are more likely to have parents who smoked, a father who worked as a manual labourer, and have moved houses during childhood. These demographic factors are used as controls in the mediation analysis — referred to as $\boldsymbol{X}_i$ in the previous section.

Figure 5: Correlation Between Ed PGI and Demographic Factors



Next, in Table 2 I show estimates a regression of the first stage ($Z \to D$) and the reduced form ($Z \to Y$) and compare these results to estimates to a standard IV approach. These estimates are using a binary version of the Ed PGI ($Z_i = 1$ if higher than the mean, 0 otherwise), and binary measure for having attended higher education or not; these transformations make these estimates comparable to the mediation analysis to follow. The IV estimates assume that Ed PGI satisfies the exclusion restiction, a dubious assumption in this setting, and I include the IV estimates for comparison purpose. Ed PGI strongly predicts university attendance; a high Ed PGI $Z_i = 1$ leads a 9.5% increase in rate of having attended university. This is after controlling for all the mentioned demographics, and mentioned other

gene scores, and an F statistic of 72 shows that the Ed PGI is strong for higher education. Similarly, Ed PGI strongly predicts annual earnings; a high score Ed PGI $Z_i = 1$ leads a 9.5% increase in annual earnings.[9] The IV estimates, using E Gene Score as an instrument for higher education, give implausibly high estimates for the returns to higher education. This illustrates the difference between IV methods and mediation methods.

Table 2: OLS + 2SLS Estimates, for Effect of Ed PGI on Education and Earnings in HRS Data.

	First-stage	Reduced-form	Naïve IV
	Outcome:		
	University, D_i	log Earnings, Y_i	
	(1)	(2)	(3)
EA Score, Z_i	0.095	0.095	
	(0.014)	(0.019)	
Attended University, D_i			1.002
			(0.212)
Constant	−2.716	7.628	10.350
	(1.603)	(2.249)	(2.427)
Demographic Controls?	Yes	Yes	Yes
Observations	5,409	5,409	5,409
R^2	0.157	0.145	0.044
F Statistic (df = 14; 5394)	71.992	65.382	

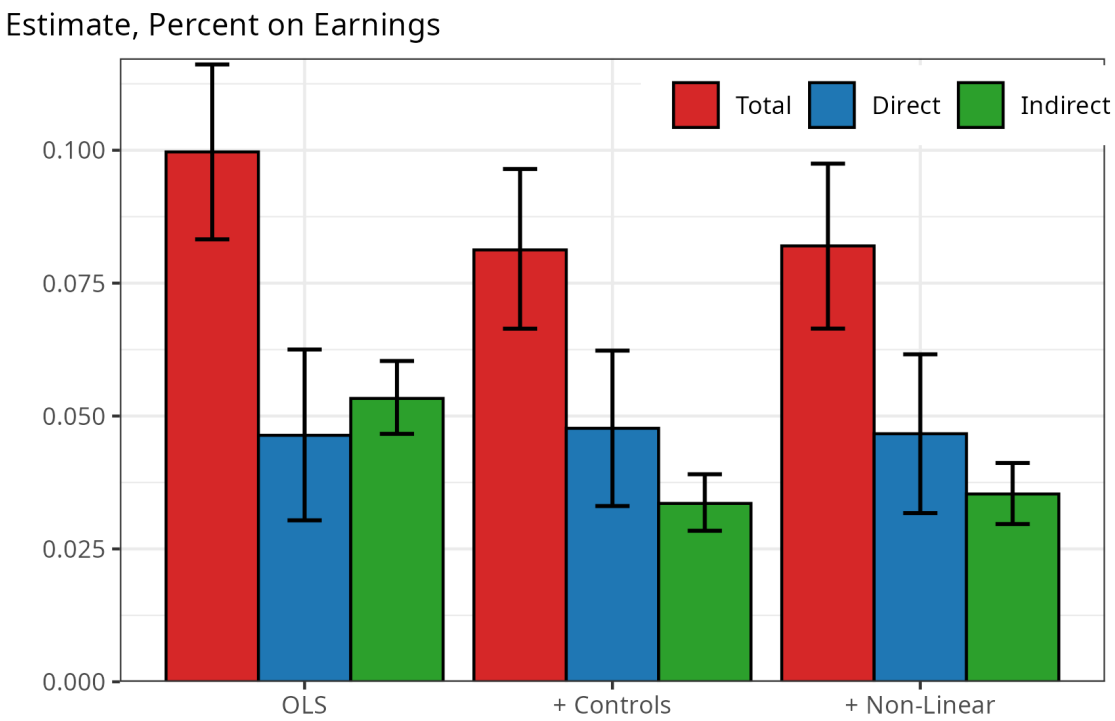
Note: This table shows OLS results of the first stage, reduced form, and 2SLS (naïvely assuming exclusion) in the HRS analysis sample.

I then move to conventional mediation methods to estimate the direct and indirect channel between Ed PGI and later-life earnings, and show the estimates in Figure 6. The red column shows the reduced form association between (binary) Ed PGI and log earnings; an increase from below mean to above mean Ed PGI is associated with an increase of 18% in later-life earning, and 15% if including demographic controls in the regression linearly or non-linearly. Without controls, the direct genetic channel is around 9% and the education channel 9%.

[9] **??** shows the corresponding table for how reduced form estimates are moderated by the demographic and other genetic controls, which shows similar modification to the first-stage.

With controls, the direct genetic channel is around 9% and the education channel 6%; the standard errors are large enough to rule out zero in either the direct or indirect channels, and the 95% confidence interval of percent mediated through education is 30–50%.

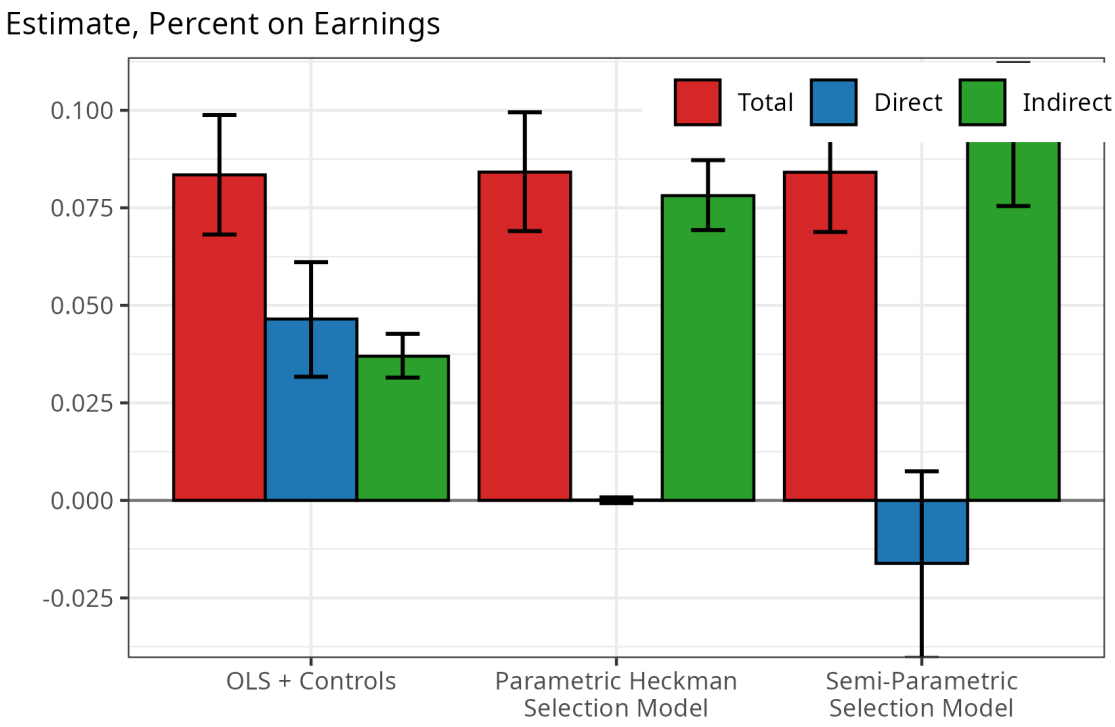
Figure 6: Direct and Indirect Estimates, using OLS.



The approach in Figure 6 assumes sequential ignorability, although the previous section brought doubt to this assumption in observational work — such as this setting. As such, I perform the exact same analysis above, but instead of estimating the regressions by OLS, I use sample selection models. The first column shows estimates of OLS with demographic controls, the second uses the same specification with a Heckman switching-selection model (Heckman 1979), and the third uses a two-stage control function estimation strategy (which is, in theory, roughly the same exercise as a sample selection model).

The selection models both estimate that the direct genetic channel is indistinguishable from zero, and assign most of the association to the indirect channel (through education).

Figure 7: Direct and Indirect Estimates, using Selection Models.



This is in contrast to the conventional mediation estimates, which estimate roughly 50-50% between the direct and indirect channels, and show the conceptual difference in modelling unobserved confounding for education in the mediation analysis.

6 Summary and Concluding Remarks

This paper studies the returns to higher education, using IV methods from the epidemiology literature and adjustments from the causal mediation literature to tackle violations of the exclusion restriction. First, I derive identification of the average mechanism effect under a selection-on-observables type assumption, and partial identification when unobserved selection confounding. I apply these methods to a sample of retirement age Americans in the years 1990–2021, using genetic information to instrument for higher education, estimating that higher education leads to roughly 40% higher earnings (point estimates), or between

8–44% higher earnings (partial bounds). Additionally, women had significantly higher returns to higher education over this time period.

The methods here provide alternatives to assuming the exclusion restriction in empirical applications of IV models, so can be useful in sensitivity analyses for any application of IV methods. Mendelian randomisation is a particularly useful application of IV methods, though the exclusion restriction is particularly problematic in practice. The approach allows researchers to use MR to study effects of both health conditions and behaviours with significant selection-into-treatment concerns, such as higher education.

References

Angrist, J. D. (1998), 'Estimating the labor market impact of voluntary military service using social security data on military applicants', *Econometrica* **66**(2), 249–288. 12

Angrist, J. D. & Pischke, J.-S. (2009), *Mostly harmless econometrics: An empiricist's companion*, Princeton university press. 18

Athey, S., Tibshirani, J. & Wager, S. (2019), 'Generalized random forests', *The Annals of Statistics* **47**(2), 1148–1178. 26

Bach, P., Chernozhukov, V., Kurz, M. S., Spindler, M. & Klaassen, S. (2024), 'DoubleML — An object-oriented implementation of double machine learning in R'. https://doi.org/10.18637/jss.v108.i03. 26

Barth, D., Papageorge, N. W. & Thom, K. (2020), 'Genetic endowments and wealth inequality', *Journal of Political Economy* **128**(4), 1474–1522. 2, 3

Carvalho, L. S. (2024), 'Genetics and socioeconomic status: Some preliminary evidence on mechanisms'. 3

Frölich, M. & Huber, M. (2017), 'Direct and indirect treatment effects–causal chains and mediation analysis with instrumental variables', *Journal of the Royal Statistical Society Series B: Statistical Methodology* **79**(5), 1645–1666. 18

Health and Retirement Study Core Public Use Dataset (2020). Produced and distributed by the University of Michigan with funding from the National Institute on Aging (grant number NIA U01AG009740). Ann Arbor, MI, (2020). 4

Health and Retirement Study Polygenic Scores – Release 4.3 (2023). Produced and distributed by the University of Michigan with funding from the National Institute on Aging (grant number NIA U01AG009740). Ann Arbor, MI, (2020). 4, 6

Heckman, J. (1974), 'Shadow prices, market wages, and labor supply', *Econometrica: journal of the econometric society* pp. 679–694. 18

Heckman, J., Ichimura, H., Smith, J. & Todd, P. (1998), 'Characterizing selection bias using experimental data', *Econometrica* **66**(5), 1017–1098. 13, 14

Heckman, J. J. (1979), 'Sample selection bias as a specification error', *Econometrica: Journal of the econometric society* pp. 153–161. 3, 18, 21

Heckman, J. J. & Vytlacil, E. (2005), 'Structural equations, treatment effects, and econometric policy evaluation 1', *Econometrica* **73**(3), 669–738. 3, 16

Hlavac, M. (2018), *stargazer: Well-Formatted Regression and Summary Statistics Tables*, Central European Labour Studies Institute (CELSI). R package version 5.2.2, https://CRAN.R-project.org/package=stargazer. 26

Houmark, M. A., Ronda, V., Agerbo, E., Mortensen, P. B. & Rosholm, M. (2022), Genetic and socioeconomic achievement gaps in elementary school, Technical report, Institute of Labor Economics (IZA). https://doi.org/10.2139/ssrn.4158731. 2

Huber, M. (2020), 'Mediation analysis', *Handbook of labor, human resources and population economics* pp. 1–38. 3

Imai, K., Keele, L. & Yamamoto, T. (2010), 'Identification, inference and sensitivity analysis for causal mediation effects', *Statistical Science* pp. 51–71. 3, 11, 12, 26

Imai, K., Tingley, D. & Yamamoto, T. (2013), 'Experimental designs for identifying causal mechanisms', *Journal of the Royal Statistical Society Series A: Statistics in Society* **176**(1), 5–51. 13

Imbens, G. & Angrist, J. (1994), 'Identification and estimation of local average treatment effects', *Econometrica* **62**(2), 467–475. 10, 15

Keele, L., Tingley, D. & Yamamoto, T. (2015), 'Identifying mechanisms behind policy interventions via causal mediation analysis', *Journal of Policy Analysis and Management* **34**(4), 937–963. 14

Lawlor, D. A., Harbord, R. M., Sterne, J. A., Timpson, N. & Davey Smith, G. (2008), 'Mendelian randomization: using genes as instruments for making causal inferences in epidemiology', *Statistics in medicine* **27**(8), 1133–1163. 8

Lee, J. J., Wedow, R., Okbay, A., Kong, E., Maghzian, O., Zacher, M., Nguyen-Viet, T. A., Bowers, P., Sidorenko, J., Karlsson Linnér, R. et al. (2018), 'Gene discovery and polygenic prediction from a genome-wide association study of educational attainment in 1.1 million individuals', *Nature genetics* **50**(8), 1112–1121. 1, 2, 7, 8, 9

Marioni, R. E., Ritchie, S. J., Joshi, P. K., Hagenaars, S. P., Okbay, A., Fischer, K., Adams, M. J., Hill, W. D., Davies, G., Consortium, S. S. G. A. et al. (2016), 'Genetic variants linked to education predict longevity', *Proceedings of the National Academy of Sciences* **113**(47), 13366–13371. https://doi.org/10.1073/pnas.1605334113. 2

Papageorge, N. W. & Thom, K. (2020), 'Genes, education, and labor market outcomes: evidence from the health and retirement study', *Journal of the European Economic Association* **18**(3), 1351–1399. 2, 3, 8, 9

R Core Team (2023), *R: A Language and Environment for Statistical Computing*, R Foundation for Statistical Computing, Vienna, Austria.
URL: *https://www.R-project.org/* 26

Robins, J. (1986), 'A new approach to causal inference in mortality studies with a sustained exposure period—application to control of the healthy worker survivor effect', *Mathematical modelling* **7**(9-12), 1393–1512. 11

Słoczyński, T. (2022), 'Interpreting ols estimands when treatment effects are heterogeneous: Smaller groups get larger weights', *Review of Economics and Statistics* **104**(3), 501–509. 12

Smith, A. (1776), *The Wealth of Nations*, Vol. 1. 1

Spiller, W., Slichter, D., Bowden, J. & Davey Smith, G. (2019), 'Detecting and correcting for bias in mendelian randomization analyses using gene-by-environment interactions', *International journal of epidemiology* **48**(3), 702–712. 3

Tibshirani, J., Athey, S., Sverdrup, E. & Wager, S. (2023), *grf: Generalized Random Forests*. R package version 2.3.0, https://CRAN.R-project.org/package=grf. 26

Uffelmann, E., Huang, Q. Q., Munung, N. S., De Vries, J., Okada, Y., Martin, A. R., Martin, H. C., Lappalainen, T. & Posthuma, D. (2021), 'Genome-wide association studies', *Nature Reviews Methods Primers* **1**(1), 59. 7

Van Kippersluis, H. & Rietveld, C. A. (2018), 'Pleiotropy-robust mendelian randomization', *International journal of epidemiology* **47**(4), 1279–1288. 3

Vytlacil, E. (2002), 'Independence, monotonicity, and latent index models: An equivalence result', *Econometrica* **70**(1), 331–341. 15, 27

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K. & Yutani, H. (2019), 'Welcome to the tidyverse', *Journal of Open Source Software* **4**(43), 1686. https://doi.org/10.21105/joss.01686. 26

A Appendix

This project used computational tools which are fully open-source. Any comments or suggestions may be sent to me at `seh325@cornell.edu`, or raised as an issue on the Github project.

A number of statistical packages, for the R language (R Core Team 2023), made the empirical analysis for this paper possible.

- *Tidyverse* (Wickham et al. 2019) collected tools for data analysis in the R language.
- *DoubleML* (Bach et al. 2024) implemented doubly robust methods used in the empirical analysis.
- *GRF* (Athey et al. 2019, Tibshirani et al. 2023) compiled forest computational tools for the R language.
- *Stargazer* (Hlavac 2018) provided methods to efficiently convert empirical results into presentable output in LaTeX.

A.1 Identification in Causal Mediation

Imai et al. (2010, Theorem 1) states that the direct and indirect effects are identified under sequential ignorability, at each level of $Z_i = 0, 1$. For $z' = 0, 1$:

$$\mathbb{E}\left[Y_i(1, D_i(z')) - Y_i(0, D_i(z'))\right] = \int\int \Big(\mathbb{E}\left[Y_i \,|\, Z_i = 1, D_i, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i \,|\, Z_i = 0, D_i, \boldsymbol{X}_i\right]\Big) dF_{D_i \,|\, Z_i = z', \boldsymbol{X}_i} dF_{\boldsymbol{X}_i},$$

$$\mathbb{E}\left[Y_i(z', D_i(1)) - Y_i(z', D_i(0))\right] = \int\int \mathbb{E}\left[Y_i \,|\, Z_i = z', D_i, \boldsymbol{X}_i\right] \Big(dF_{D_i \,|\, Z_i = 1, \boldsymbol{X}_i} - dF_{D_i \,|\, Z_i = 0, \boldsymbol{X}_i}\Big) dF_{\boldsymbol{X}_i}.$$

I focus on the averages, which are identified by consequence of the above.

$$\mathbb{E}\left[Y_i(1, D_i(Z_i)) - Y_i(0, D_i(Z_i))\right] = \mathbb{E}_{Z_i}\left[\mathbb{E}\left[Y_i(1, D_i(z')) - Y_i(0, D_i(z')) \,|\, Z_i = z'\right]\right]$$
$$\mathbb{E}\left[Y_i(Z_i, D_i(1)) - Y_i(Z_i, D_i(0))\right] = \mathbb{E}_{Z_i}\left[\mathbb{E}\left[Y_i(z', D_i(1)) - Y_i(z', D_i(0)) \,|\, Z_i = z'\right]\right]$$

My estimand for the average direct effect is a simple rearrangement of the above. The estimand for the average indirect effect relies on a different sequence, relying on (1) sequential ignorability, (2) conditional monotonicity. These give (1) identification of, and equivalence between, LADE conditional on $\boldsymbol{X}_i$ and ADE conditional on $\boldsymbol{X}_i$, (2) identification of the complier score.

$$
\begin{aligned}
&\mathbb{E}\left[Y_i(Z_i, D_i(1)) - Y_i(Z_i, D_i(0)) \mid \boldsymbol{X}_i\right] \\
&= \Pr\left(D_i(1) = 1, D_i(0) = 0 \mid \boldsymbol{X}_i\right) \mathbb{E}\left[Y_i(Z_i, 1) - Y_i(Z_i, 0) \mid D_i(1) = 1, D_i(0) = 0, \boldsymbol{X}_i\right] \\
&= \Pr\left(D_i(1) = 1, D_i(0) = 0 \mid \boldsymbol{X}_i\right) \mathbb{E}\left[Y_i(Z_i, 1) - Y_i(Z_i, 0) \mid \boldsymbol{X}_i\right] \\
&= \Big(\mathbb{E}\left[D_i \mid Z_i = 1, \boldsymbol{X}_i\right] - \mathbb{E}\left[D_i \mid Z_i = 0, \boldsymbol{X}_i\right]\Big) \mathbb{E}\left[Y_i(Z_i, 1) - Y_i(Z_i, 0) \mid \boldsymbol{X}_i\right] \\
&= \Big(\mathbb{E}\left[D_i \mid Z_i = 1, \boldsymbol{X}_i\right] - \mathbb{E}\left[D_i \mid Z_i = 0, \boldsymbol{X}_i\right]\Big)\Big(\mathbb{E}\left[Y_i \mid Z_i, D_i = 1, \boldsymbol{X}_i\right] - \mathbb{E}\left[Y_i \mid Z_i, D_i = 0, \boldsymbol{X}_i\right]\Big)
\end{aligned}
$$

Monotonicity is not technically required for the above. Breaking monotonicity would not change the identification of any of the above; it would be the same except replacing the complier score with a complier or defier score, $\Pr\left(D_i(1) \neq D_i(0) \mid \boldsymbol{X}_i\right) = \mathbb{E}\left[D_i \mid Z_i = 1, \boldsymbol{X}_i\right] - \mathbb{E}\left[D_i \mid Z_i = 0, \boldsymbol{X}_i\right]$.

A.2 Continuous Average Causal Responses

Section here relating the approach to the average causal response function (see e.g., Angrist Imbens JASA 1996, Andrew Bacon for DiD 2023).

A.3 Previous Literature

Create a table in this section that surveys previous research which employs mediation methods while having a clear causal design for Z_i, but not D_i.

Paper	Field	Research Design for Z_i	Research Design for D_i	Selection bias?
Paper name 1.				

A.4 Selection Bias in Mediation Estimates

Write the proof in here. It is long....

A.5 Proof of the Selection Model Representation

Write the proof in here, following Vytlacil (2002) construction in the forward direction. Note that the notation needs updating for no exclusion restriction.

A.6 A Regression Framework for Direct and Indirect Effects

Put $\mu_D(Z; \boldsymbol{X}) = \mathbb{E}\left[Y_i(Z, D) \mid \boldsymbol{X}\right]$, so we have the following expressions.

$$
Y_i(Z_i, 0) = \mu_0(Z_i; \boldsymbol{X}_i) + U_{0,i}, \quad Y_i(Z_i, 1) = \mu_1(Z_i; \boldsymbol{X}_i) + U_{1,i}
$$

$U_{0,i}, U_{1,i}$ are error terms with unknown distributions, mean independent of $Z_i, \boldsymbol{X}_i$ by definition — but possibly correlated with D_i.

Z_i is assumed randomly assigned, independent of potential outcomes, so that $U_{0,i}, U_{1,i} \perp\!\!\!\perp Z_i$. Thus, the first-stage $Z \to Y$ leads to unbiased estimates.

$$\begin{aligned}
D_i &= Z_i D_i(1) + (1 - Z_i) D_i(0) \\
&= D_i(0) + Z_i \left[D_i(1) - D_i(0)\right] \\
&= \underbrace{\mathbb{E}\left[D_i(0)\right]}_{\text{Intercept}} + \underbrace{Z_i \mathbb{E}\left[D_i(1) - D_i(0)\right]}_{\text{Regressor}} \\
&\quad + \underbrace{D_i(0) - \mathbb{E}\left[D_i(0)\right] + Z_i\big(D_i(1) - D_i(0) - \mathbb{E}\left[D_i(1) - D_i(0)\right]\big)}_{\text{Mean-zero, independent error term}} \\
&=: \phi_i + \pi_i Z_i + \eta_i \\
\implies \mathbb{E}\left[D_i \,|\, Z_i\right] &= \mathbb{E}\left[\phi_i\right] + \mathbb{E}\left[\phi_i\right] Z_i + \mathbb{E}\left[\eta_i\right], \text{ and thus unbiased estimates since } Z_i \perp\!\!\!\perp \phi_i, \eta_i.
\end{aligned}$$

Z_i is also assumed independent of potential outcomes $Y_i(,.,)$, so that $U_{0,i}, U_{1,i} \perp\!\!\!\perp Z_i$. Thus, the reduced form regression $Z \to Y$ also leads to unbiased estimates.

The same cannot be said of the regression that estimates direct and indirect effects, without further assumptions.

$$\begin{aligned}
Y_i &= Z_i Y_i(1, D_i(1)) + (1 - Z_i) Y_i(0, D_i(0)) \\
&= Z_i D_i Y_i(1,1) \\
&\quad + (1 - Z_i) D_i Y_i(0,1) \\
&\quad + Z_i (1 - D_i) Y_i(1,0) \\
&\quad + (1 - Z_i)(1 - D_i) Y_i(0,0) \\
&= Y_i(0,0) \\
&\quad + Z_i \left[Y_i(1,0) - Y_i(0,0)\right] \\
&\quad + D_i \left[Y_i(0,1) - Y_i(0,0)\right] \\
&\quad + Z_i D_i \left[Y_i(1,1) - Y_i(1,0) - (Y_i(0,1) - Y_i(0,0))\right]
\end{aligned}$$

And so Y_i can be written as a regression equation in terms of the observed factors and error terms.

$$\begin{aligned}
\mathbb{E}\left[Y_i \,|\, Z_i, D_i, \boldsymbol{X}_i\right] &= \mu_0(0; \boldsymbol{X}_i) \\
&\quad + Z_i \left[\mu_0(1; \boldsymbol{X}_i) - \mu_0(0; \boldsymbol{X}_i)\right] \\
&\quad + D_i \left[\mu_1(0; \boldsymbol{X}_i) - \mu_0(0; \boldsymbol{X}_i)\right] \\
&\quad + Z_i D_i \left[\mu_1(1; \boldsymbol{X}_i) - \mu_0(1; \boldsymbol{X}_i) - (\mu_1(0; \boldsymbol{X}_i) - \mu_0(0; \boldsymbol{X}_i))\right] \\
&\quad + U_{0,i} + D_i \left(U_{1,i} - U_{0,i}\right) \\
&=: \alpha_i + \beta_i D_i + \gamma_i Z_i + \delta_i Z_i D_i + \underbrace{U_{0,i} + D_i \left(U_{1,i} - U_{0,i}\right)}_{=\varepsilon_i}
\end{aligned}$$

$\alpha_i, \beta_i, \delta_i$ are the relevant direct effect under $D_i = 1$, indirect effect under $Z_i = 1$, δ_i the interaction effect, and ε_i the remaining error term. Collecting for the expressions of the direct and indirect effects:[10]

$$\mathbb{E}\left[Y_i(Z_i, D_i(1)) - Y_i(Z_i, D_i(0))\right] = \mathbb{E}\left[\pi_i\left(\beta_i + Z_i\delta_i\right)\right]$$
$$\mathbb{E}\left[Y_i(1, D_i(Z_i)) - Y_i(0, D_i(Z_i))\right] = \mathbb{E}\left[\gamma_i + \delta_i D_i\right]$$

If sequential ignorability does not hold, then the regression estimates from estimating the mediation equations (without adjusting for the contaminated bias term) suffer from omitted variables bias:

$$\mathbb{E}\left[\widehat{\alpha}\right] = \mathbb{E}\left[\alpha_i\right] + \mathbb{E}\left[D_i\left(U_{1,i} - U_{0,i}\right)\right]$$
$$\mathbb{E}\left[\widehat{\beta}\right] = \mathbb{E}\left[\beta_i\right] + \frac{\mathrm{Cov}\left(D_i,\ D_i\left(U_{1,i} - U_{0,i}\right)\right)}{\mathrm{Var}\left(D_i\right)}$$
$$\mathbb{E}\left[\widehat{\gamma}\right] = \mathbb{E}\left[\gamma_i\right] + \frac{\mathrm{Cov}\left(Z_i,\ D_i\left(U_{1,i} - U_{0,i}\right)\right)}{\mathrm{Var}\left(Z_i\right)}$$
$$\mathbb{E}\left[\widehat{\delta}\right] = \mathbb{E}\left[\delta_i\right] + \frac{\mathrm{Cov}\left(Z_iD_i,\ D_i\left(U_{1,i} - U_{0,i}\right)\right)}{\mathrm{Var}\left(Z_iD_i\right)}$$

And so the direct and indirect effect estimates are contaminated by these bias terms.

[10] These equations have simpler expressions after assuming constant treatment effects in a linear framework; I have avoided this as having compliers, and controlling for observed factors $\boldsymbol{X}_i$ only makes sense in the case of heterogeneous treatment effects.